

The Subgaussian-Diameter Concentration Constant Is Sharp

Counterexample packet for arXiv:1309.1007

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Status: likely valid full negative answer to the constant-recovery question, with a sharpness theorem. The sharp McDiarmid coefficient 2 cannot survive when ordinary diameters are replaced by subgaussian diameters. In fact the coefficient $1/2$ in Theorem 1 of the source is the largest possible universal coefficient.

1 Source and exact question

The source is A. Kontorovich, *Concentration in unbounded metric spaces and algorithmic stability*, arXiv:1309.1007 (2013), later published in Proceedings of ICML 2014 [1]. Its Theorem 1 states that for a 1-Lipschitz function on an ℓ_1 product,

$$\mathbb{P}\{|\varphi - \mathbb{E}\varphi| > t\} \leq 2 \exp\left(-\frac{t^2}{2 \sum_i \Delta_{\text{SG}}(\mathcal{X}_i)^2}\right). \quad (1)$$

Section 8 asks how tight this is, noting the coefficient $1/2$ in (1) versus McDiarmid's optimal coefficient 2, and asks for a bound recovering McDiarmid's in the finite-diameter case. The proceedings version states the issue explicitly: the author did not know whether McDiarmid's inequality holds verbatim with $\text{diam}(\mathcal{X}_i)$ replaced by $\Delta_{\text{SG}}(\mathcal{X}_i)$.

We have given a concentration inequality for metric spaces with unbounded diameter, showed its applicability to algorithmic stability with unbounded losses, and gave an extension to non-independent sampling processes. Some fascinating questions remain:

- (i) How tight is Theorem 1? First there is the vexing matter of having a worse constant in the exponent (i.e., $1/2$) than McDiarmid's (optimal) constant 2. Although this gap is not of critical importance, one would like a bound that recovers McDiarmid's in the finite-diameter case. More importantly, is it the case that finite subgaussian diameter is necessary for subgaussian concentration of all Lipschitz functions? That is, given the metric probability spaces $(\mathcal{X}_i, \rho_i, \mu_i)$, $i \in [n]$, can one always exhibit a 1-Lipschitz $\varphi : \mathcal{X}^n \rightarrow \mathbb{R}$ that achieves a nearly matching lower bound?

Figure 1: The tightness question in Section 8, page 14 of the arXiv source.

2 Definitions and proposed strengthening

For independent $X, X' \sim \mu$ and an independent Rademacher sign ε , the source defines

$$\Xi(\mathcal{X}) = \varepsilon \rho(X, X'), \quad \Delta_{\text{SG}}(\mathcal{X}) = \sigma^*(\Xi), \quad \sigma^*(Z)^2 = \sup_{s \neq 0} \frac{2 \log \mathbb{E} e^{sZ}}{s^2}.$$

For a universal coefficient $c > 0$, let $C(c)$ denote

$$\mathbb{P}\{|\varphi - \mathbb{E}\varphi| > t\} \leq 2 \exp\left(-\frac{ct^2}{\sum_i \Delta_{\text{SG}}(\mathcal{X}_i)^2}\right) \quad (2)$$

for all such products and all 1-Lipschitz φ . The source proves $C(1/2)$. Verbatim McDiarmid recovery would be $C(2)$.

3 Proof intuition

Use one coordinate with the two-point metric and a Bernoulli law. The signed distance is the difference of two independent Bernoulli variables. For the uniform law its subgaussian diameter is exactly $1/\sqrt{2}$, already too small for coefficient 2 to control the two atoms of a 1-Lipschitz function.

More sharply, as the success probability p tends to zero, the optimal subgaussian variance of the signed distance is asymptotic to $1/(2\log(1/p))$. The coordinate function still has a deviation nearly 1 with probability p . Substitution into (2) forces $p \lesssim p^{2c}$, impossible for every $c > 1/2$. Thus the source constant is universally sharp.

4 An exact two-point counterexample

Theorem 1 (McDiarmid's coefficient fails). *Let $\mathcal{X} = \{0, 1\}$, let $\rho(x, y) = |x - y|$, and give both points mass $1/2$. For $\varphi(x) = x$,*

$$\Delta_{\text{SG}}(\mathcal{X})^2 = \frac{1}{2}.$$

At $t = 9/20$,

$$\mathbb{P}\{|\varphi - \mathbb{E}\varphi| > t\} = 1 \quad \text{but} \quad 2 \exp\left(-\frac{2t^2}{\Delta_{\text{SG}}(\mathcal{X})^2}\right) = 2e^{-81/100} < 1.$$

Consequently $C(2)$ is false, already for one finite two-point space.

Proof. If B, B' are independent uniform Bernoulli variables, then $\Xi = \varepsilon|B - B'|$ has moment-generating function

$$M(s) = \frac{1}{2} + \frac{1}{2} \cosh s = \cosh^2(s/2).$$

Hence, with $u = s/2$,

$$\Delta_{\text{SG}}(\mathcal{X})^2 = \sup_{s \neq 0} \frac{2 \log M(s)}{s^2} = \sup_{u \neq 0} \frac{\log \cosh u}{u^2} = \frac{1}{2}.$$

Indeed $\log \cosh u \leq u^2/2$, with equality approached as $u \rightarrow 0$. Also $\mathbb{E}\varphi = 1/2$, so $|\varphi - \mathbb{E}\varphi| = 1/2$ almost surely. $\square$

5 Sharpness of the source coefficient

Lemma 1 (Rare-Bernoulli diameter). *Give 1 mass p and 0 mass $1 - p$ on the same two-point metric space, and write $\Delta_p = \Delta_{\text{SG}}(\mathcal{X}, \rho, \mu_p)$. Then*

$$\Delta_p^2 \sim \frac{1}{2 \log(1/p)} \quad (p \downarrow 0).$$

Proof. Put $L = \log(1/p)$ and $q = 2p(1-p)$. The signed distance has the same law as $B - B'$, for independent Bernoulli(p) variables, so

$$M_p(s) = 1 + q(\cosh s - 1), \quad \Delta_p^2 = \sup_{s>0} \frac{2 \log M_p(s)}{s^2}. \quad (3)$$

At $s = 2L$,

$$M_p(2L) = 1 + 2p(1-p)(\cosh(2L) - 1) = p^{-1} + O(p),$$

and hence $\Delta_p^2 \geq (1 + o(1))/(2L)$.

For the reverse estimate, assume L is large. If $0 < s \leq L/2$, then

$$\log M_p(s) \leq q(\cosh s - 1) \leq \frac{qs^2 e^s}{2},$$

so the quotient in (3) is at most $2e^{-L/2}$. If $L/2 \leq s \leq L$, then $M_p(s) \leq 1 + 2e^{s-L} \leq 3$, giving $8 \log(3)/L^2$. Finally, if $s \geq L$, then

$$\log M_p(s) \leq s - L + \log 3.$$

The maximum of $2(s - L + \log 3)/s^2$ on this range is at most $1/[2(L - \log 3)]$. The maximum of these three estimates is $(1 + o(1))/(2L)$. $\square$

Theorem 2 (Universal optimality). *The inequality $C(c)$ holds universally if and only if $c \leq 1/2$. Thus the coefficient $1/2$ in Theorem 1 of the source is optimal.*

Proof. The source proves $C(1/2)$, and smaller coefficients follow. Fix $c > 1/2$ and use Lemma 1 with $\varphi(x) = x$. For $t_p = 1 - 2p$ and all small p ,

$$\mathbb{P}\{|\varphi - \mathbb{E}\varphi| > t_p\} = p.$$

If $C(c)$ held, then

$$p \leq 2 \exp\left(-\frac{c(1-2p)^2}{\Delta_p^2}\right) = 2p^{2c+o(1)}.$$

After division by p , the right side tends to zero since $2c - 1 > 0$, a contradiction. $\square$

6 Companion answers to nearby source questions

Theorem 3 (Equality of the two diameters). *Let a metric probability space have finite diameter $D > 0$, and set $R = \rho(X, X')$. Then*

$$\Delta_{\text{SG}}(\mathcal{X}) = D \iff R = D \text{ almost surely.}$$

Consequently $\Delta_{\text{SG}}(\mathcal{X}) < D$ for every nontrivial separable metric probability space, and hence for every Polish, countably supported, or Radon metric probability space.

Proof. Since $\mathbb{E}e^{s\Xi} = \mathbb{E} \cosh(sR)$, set

$$h(s) = \frac{2 \log \mathbb{E} \cosh(sR)}{s^2} \quad (s \neq 0), \quad h(0) = \mathbb{E}R^2.$$

This is continuous at zero and tends to zero as $|s| \rightarrow \infty$. If $\mathbb{P}\{R < D\} > 0$, then $h(0) < D^2$ and, for $s \neq 0$,

$$\mathbb{E} \cosh(sR) < \cosh(sD) < e^{s^2 D^2/2}.$$

Continuity and decay give $\sup_s h(s) < D^2$. Conversely, if $R = D$ almost surely, then $\text{Var}(\Xi) = D^2$, so every subgaussian proxy is at least D ; the source’s Hoeffding bound gives the reverse inequality.

If equality held on a separable space, every set of diameter less than D would be null. A countable cover by balls of radius $D/3$ is then impossible. A Radon probability is concentrated on a separable union of compact sets, reducing to this case. $\square$

Corollary 1 (Qualitative necessity). *If, for some x_0 , the 1-Lipschitz random variable $x \mapsto \rho(x, x_0)$ has a subgaussian centered law, then $\Delta_{\text{SG}}(\mathcal{X}) < \infty$. Thus subgaussianity of all Lipschitz functions does imply finite subgaussian diameter; a nearly matching quantitative lower bound remains separate.*

Proof. Put $A = \rho(X, x_0)$. Subgaussianity of $A - \mathbb{E}A$ implies $\mathbb{E}e^{A^2/K^2} < \infty$ for some K . Since $R \leq A + A'$, also $\mathbb{E}e^{R^2/K_1^2} < \infty$ for some K_1 . Enlarge K_1 so this expectation is at most 2. For $Z = \varepsilon R$,

$$\mathbb{E}|Z|^{2k} \leq 2k! K_1^{2k}.$$

As $(2k)! \geq (k!)^2$, the even moment series gives

$$\mathbb{E}e^{sZ} \leq 1 + 2 \sum_{k \geq 1} \frac{(s^2 K_1^2)^k}{k!} = 2e^{s^2 K_1^2} - 1 \leq e^{2s^2 K_1^2}.$$

Thus Z is subgaussian. $\square$

7 Verification, novelty, and limitations

Checks. The explicit example uses the exact identity $\frac{1}{2}(1 + \cosh s) = \cosh^2(s/2)$. The sharpness proof preserves the strict tail convention by taking $t_p = 1 - 2p < 1 - p$. The three-range estimate is uniform in $s > 0$. The included script is regression-only.

Bounded novelty search. The run’s four lightweight indexes were searched for arXiv:1309.1007, subgaussian diameter, McDiarmid replacement, Bernoulli sharpness, and the factor-four gap. Exact-pharse and core-keyword arXiv/web searches found the source and classical work on Bernoulli subgaussian constants, but no direct answer to the replacement question and no statement of Theorem 2. Novelty is plausible but remains subject to expert review.

Scope and review. Theorem 2 completely resolves the universal constant question. The packet does not produce a nearly matching lower-bound Lipschitz function for every product space, address the Kutin–Niyogi metric conjecture, or prove stability of the listed algorithms. In unrestricted nonseparable, non-Radon spaces, Theorem 3 is a characterization, not a ZFC existence claim. Human review should check the uniform upper estimate in Lemma 1 and the proceedings version’s interpretation of verbatim replacement.

References

- [1] A. Kontorovich, *Concentration in unbounded metric spaces and algorithmic stability*, Proceedings of the 31st International Conference on Machine Learning, PMLR **32** (2014), 28–36; [arXiv:1309.1007](#).